

2024

(Session : 2023-26)

(Paper ID : 11001)

Time : 3 hours

Full Marks : 80

Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. Find the Eigen values and Eigen vectors of the following matrix:

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

2. Using matrix method, solve the following system of equations:

$$\begin{aligned} x + y + z &= 6 \\ 2x + y - 3z &= -5 \\ 3x - 2y + z &= 2 \end{aligned}$$

3. If $y^{1/m} + y^{-1/m} = 2x$, prove that

$$(x^2 - 1) \cdot y_2 + xy_1 - m^2y = 0$$

4. Show that $\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \dots$ using Maclaurin's series.

5. Evaluate $I = \int \frac{1}{2 \cos x - 3 \sin x + 5} dx$

6. Evaluate $I = \int \frac{\sin x \cos x}{1 + \sin^4 x} dx$

7. Evaluate $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1 + \sqrt{\tan x}}$

8. Solve $\frac{d^2y}{dx^2} + y \tan x = y^2 \sec x$

9. Solve $(x^2 - yx^2)dy + (y^2 - xy^2)dx = 0$

10. Write short notes on following:

- (a) Inverse matrix
- (b) Successive Differentiation
- (c) Transpose of matrix
- (d) Singular matrix

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